

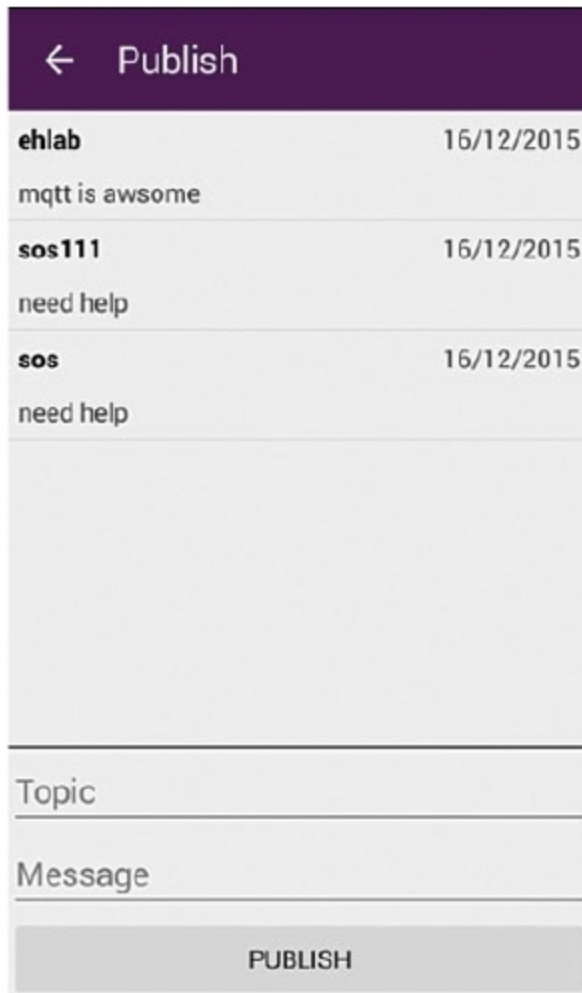


Fig. 5: Create a topic on phone1 to publish

MQTT has been successfully connected (Fig. 3).

4. Create a topic in phone1 from the page, as shown in Fig. 4.

On the top-right corner of the application, press Upload near Settings to publish the topic. You will see a page as shown in Fig. 5.

In Fig. 5, there are two text boxes: topic and message.

Create a topic as `sos111`, and the message as `need help`. Press Publish on phone1.

Next, subscribe to `sos111` in phone2. For this, go back to the application and in Add Topic text box, as shown in Fig. 4, type topic name as `sos111`, same as in phone1. You will get need help message on phone2.

Setting up MQTT broker on Raspberry Pi

You can set up your own MQTT server/broker on any cloud server, a local machine or Raspberry Pi. In this section we learn how to set up a local MQTT server on Raspberry Pi. Raspberry Pi 2 board with an Internet connection was used during testing. Follow the steps given below.

Install some dependencies in Raspberry Pi before compiling and running MQTT server. For that, run the following commands on the terminal:

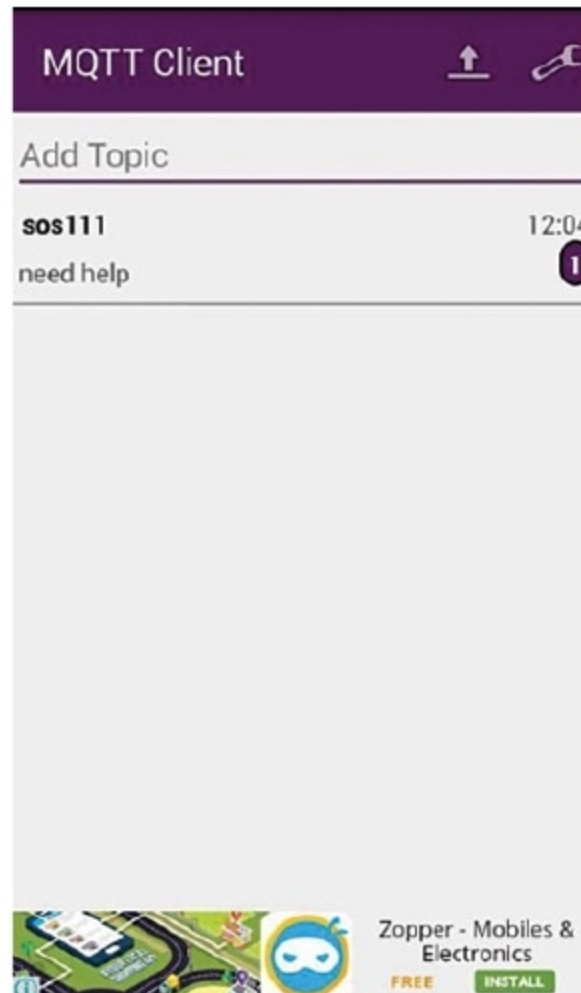


Fig. 6: Subscribe to topic on phone2

```
$ sudo apt-get update
$ sudo apt-get install libssl-dev
$ sudo apt-get install cmake
$ sudo apt-get install libc-ares-dev
$ sudo apt-get install uuid-dev
$ sudo apt-get install daemon
```

Next, download `libwebsockets` using the command given below.

```
$ git clone https://libwebsockets.org/
repo/libwebsockets
```

Unpack it using the following command:

```
$ tar zxvf libwebsockets
```

Change directory using the following command:

```
$ cd libwebsockets
```

Make a build directory inside `libwebsockets` using the commands given below.

```
$ mkdir build
$ cd build
```

Then, to build, the command is:

```
$ cmake .. # (note the ..)
```

Run the following command to install:

```
$ sudo make install
```

Next, rebuild the library cache using the following command:

```
$ sudo ldconfig
```

Change the directory to Home using the following command:

```
$ cd
```

Download the source code of MQTT broker, Mosquitto, using the

following command:

```
$ wget http://mosquitto.org/files/source/
mosquitto-1.4.1.tar.gz
```

Unpack it using the following command:

```
$ tar zxvf mosquitto-1.4.1.tar.gz
$ cd mosquitto-1.4.1
```

Open and edit `config.mk` from any text editor using the following command:

```
$ cd config.mk
```

Change `WITH_WEBSOCKETS: = no` to `WITH_WEBSOCKETS: = yes`

Then, go back to Home and compile using the following command:

```
$ make
```

Install it using the following command:

```
$ sudo make install
```

Then, create a config directory inside `/etc` folder using the following command:

```
$ sudo mkdir /etc/mosquitto
```

Copy default config file to `/etc` folder using the following command:

```
$ sudo cp mosquitto.conf /etc/mosquitto
```

Add the following two lines to `/etc/mosquitto/mosquitto.conf` at the end of the file:

```
listener 9001
protocol websockets
```

Next, add a user for mosquitto using the following command:

```
$ sudo adduser mosquitto
```

Reboot and login with the username as `mosquitto`. After successful login, open the terminal window on Raspberry Pi and run following command:

```
$ mosquitto_sub -h localhost -t test
```

Open another terminal window on same Raspberry Pi and run the following command:

```
$ mosquitto_pub -h localhost -t test -m
"hello world"
```

You will see hello world message on the first terminal window.

Congrats, your first MQTT server is up and running! You can now communicate with any other device using MQTT client installed on your Android phone. **EFY**

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